

Project: Transparent Multi-Objective Food Recommendations

Focus: Human-Computer Interaction (HCI), Sustainable UX, and Explainable AI (XAI).

1. Project Overview

The objective is to design and prototype a web platform that recommends food products based on a dual-scoring system: **Nutritional Value** (Health) and **Ecological Footprint** (Carbon, Water, Land use). The core HCI challenge is managing "trade-off" visualizations—where a healthy product might have a high environmental cost (e.g., certain nuts or out-of-season fruits).

2. HCI Research & Discovery Phase

Students should begin with a rigorous user-centered design (UCD) approach:

- **Persona Development:** Create personas with conflicting goals (e.g., "The Budget-Conscious Athlete" vs. "The Zero-Waste Parent").
- **Cognitive Task Analysis:** Map how users currently evaluate products at a grocery store vs. online.
- **Ethical Inquiry:** Research the "value-action gap"—why people *want* to eat sustainably but often don't—and how interface nudges can bridge this.

3. Core Features & System Architecture

- **Multi-Criteria Search:** Users filter by nutrients (high protein, low sodium) and eco-metrics (low CO2, plastic-free packaging).
- **The "Alternative Engine":** A recommendation system that suggests a "Better For You" (Nutritional) and "Better For Earth" (Ecological) alternative for any searched item.
- **Personalization API:** A profile setting where users weigh their priorities (e.g., "I care 70% about health and 30% about the environment").

4. Explainable AI (XAI) Integration

Instead of a black-box score, the system must explain *why* it recommended a product.

- **Feature Attribution (Local Explanations):** If the system recommends oat milk over almond milk, it should display a breakdown: *"Recommended because it uses 80% less water, despite having slightly lower protein content."*
- **Contrastive Explanations:** "Why not Product B?" The UI should allow users to compare the recommendation against their usual choice, highlighting the delta in both health and eco-impact.

- **Visual XAI:** Use interactive charts (e.g., Radar/Spider charts) to show how a product performs across 5-6 dimensions simultaneously.

5. Implementation & Prototyping

- **Frontend:** Streamlit, Plotly Dash, Reflex, etc.
- **Data Sources:** Open Food Facts API (Nutrition), etc.

6. Evaluation Metrics

Students must conduct a formal user study including:

- **System Usability Scale (SUS):** To measure general ease of use.
- **Explanation Satisfaction Scale:** A specific XAI metric to see if the AI's reasoning actually increased user trust.

Note: It is important to emphasize that all the points outlined in this proposal—from the choice of frameworks to the evaluation metrics—should be viewed as **strategic recommendations** rather than rigid requirements. The intent is to provide a foundational structure that ensures the academic rigor expected at a graduate level, while allowing students the creative autonomy to adapt, expand, or modify these components based on their own user research findings and the technical constraints of their development environment.